Synoptic Question 7: The Physics of a Solar Hot Water Heater

Reading Passage

Read the following article and then answer the questions that follow.

A solar hot water heater is a device that captures solar energy and uses it to heat water for domestic use. A common design consists of a solar collector panel and an insulated storage tank. The collector panel is a flat box with a transparent glass cover and a black absorber plate. Water, or a heat-transfer fluid, circulates through pipes attached to the absorber plate, absorbing thermal energy.

This heated fluid is then stored in a large, well-insulated hot water tank. The insulation is crucial for minimizing heat loss to the cooler surrounding environment, especially overnight. The rate at which heat is lost depends on the thermal conductivity of the insulation, its thickness, the surface area of the tank, and the temperature difference between the water and the surroundings.

The aim of the system is to collect enough energy during the day to provide hot water as needed, raising its temperature from the cold-water supply temperature to a usable hot temperature, often around 60 °C or higher. The performance of the system depends on the intensity of the solar radiation, the efficiency of the collector, and the effectiveness of the tank's insulation.

Data:

A new solar heating system is being tested.

Table 1: Data for Solar Heater System

Parameter	**Value**
Area of solar collector panel, A	4.0 m^2
Average solar radiation intensity, I	750 W m^{-2}
Efficiency of solar collector	60%
Mass of water in storage tank, m_w	200 kg
Initial temperature of water in tank	15.0 °C
Ambient air temperature (outside), T_{air}	10.0 °C
Surface area of hot water tank, A_{tank}	2.5 m^2
Thickness of polyurethane insulation, d	5.0 cm
Thermal conductivity of insulation, k	0.025 W m^{-1} K^{-1}
Specific heat capacity of water, c_w	4200 J kg^{-1} K^{-1}
Specific latent heat of fusion of ice, L_f	3.34×10^5 J kg^{-1}
Specific latent heat of vaporization of water, L_v	2.26×10^6 J kg^{-1}

Questions

(a) The system is tested for a period of 5.0 hours on a sunny day.

  (i) Calculate the total power of the solar radiation incident on the collector panel.

  (ii) Calculate the total energy incident on the panel during the 5.0-hour test.

(b) Taking the collector's efficiency into account, calculate the total thermal energy Q_{gain} transferred to the water in the tank during the 5.0-hour test.

(c) Ignoring any heat losses from the tank, calculate the final temperature of the water in the tank after the 5.0-hour test.

(d) The insulation is designed to reduce heat loss.

  (i) Convert the insulation thickness of 5.0 cm into SI units.

  (ii) Explain in terms of molecular motion how heat is transferred by conduction through the insulation.

(e) Assume the water in the tank is at a constant average temperature of 50.0 °C during the test.

  (i) Calculate the temperature difference, ΔT, between the water in the tank and the ambient air.

(ii) Using the formula $P_{loss} = \frac{kA\Delta T}{d}$, calculate the initial rate of heat loss by conduction through the tank's insulation.

(f) Calculate the total energy Q_{loss} lost by conduction over the 5.0-hour period, assuming the constant average water temperature of 50.0 °C.

(g) Using your answers from (b) and (f), calculate the net energy gained by the water and determine a more realistic final temperature.

(h) Explain why the absorber plate in the collector is painted black.

(i) To test the hot water, a 1.2 kg sample of the water at the final temperature calculated in (g) is mixed with 0.50 kg of ice at 0 °C. Determine if all the ice melts. Show your working.

(j) If the heating system malfunctioned and continued to heat the 200 kg of water from 100 °C, calculate the energy required to turn 10% of the water into steam at 100 °C.

Solutions

(a) (i) Incident Power = Intensity × Area = $I \times A = (750) \times (4.0) = 3000$ W.

(ii) Time $t = 5.0$ h $\times$ 3600 s/h = 18000 s.
Incident Energy = Power × Time = $3000 \times 18000 = 5.40 \times 10^7$ J.

(b) Q_{gain} = Incident Energy × Efficiency = $(5.40 \times 10^7) \times 0.60 = 3.24 \times 10^7$ J.

(c) Using $Q = mc\Delta T$, where $\Delta T = T_f - T_i$.
$3.24 \times 10^7 = (200)(4200)(T_f - 15.0)$
$T_f - 15.0 = \frac{3.24\times10^7}{840000} = 38.57$
$T_f = 38.57 + 15.0 = 53.57$ °C.

(d) (i) $d = 5.0$ cm = 0.050 m.

(ii) Molecules in the warmer, inner layer of the insulation vibrate with higher kinetic energy. They collide with adjacent, cooler molecules, transferring kinetic energy to them. This process repeats in a chain, transferring thermal energy from the hot side to the cold side without any net movement of the material.

(e) (i) $\Delta T = T_{water} - T_{air} = 50.0 - 10.0 = 40.0$ °C (or 40.0 K).

(ii) $P_{loss} = \frac{kA_{tank}\Delta T}{d} = \frac{(0.025)(2.5)(40.0)}{0.050} = 50.0$ W.

(f) $Q_{loss} = P_{loss} \times t = (50.0) \times (18000) = 9.0 \times 10^5$ J.

(g) Net Energy Gained = $Q_{gain} - Q_{loss} = (3.24 \times 10^7) - (9.0 \times 10^5) = 3.15 \times 10^7$ J.
Using $Q_{net} = mc\Delta T$:
$3.15 \times 10^7 = (200)(4200)(T_f - 15.0)$
$T_f - 15.0 = \frac{3.15\times10^7}{840000} = 37.5$
$T_f = 37.5 + 15.0 = 52.5$ °C.

(h) Black surfaces are excellent absorbers (and emitters) of thermal radiation. The plate is painted black to absorb the maximum amount of energy from the solar radiation that passes through the glass cover.

(i) Final water temperature $T_f = 52.5$ °C.
Energy available from hot water cooling to 0 °C:
$Q_{water} = m_w c_w \Delta T = (1.2)(4200)(52.5 - 0) = 2.646 \times 10^5$ J.
Energy needed to melt all the ice:
$Q_{ice} = m_{ice} L_f = (0.50)(3.34 \times 10^5) = 1.67 \times 10^5$ J.
Since $Q_{water}(2.65 \times 10^5$ J$) > Q_{ice}(1.67 \times 10^5$ J$)$, all the ice will melt.

(j) Mass of water to be vaporized = 10% × 200 kg = 20.0 kg.
Energy required = $Q = mL_v = (20.0)(2.26 \times 10^6) = 4.52 \times 10^7$ J.

[Diagram of a solar hot water heater system with collector and tank]

Figure 1: Solar Hot Water Heater System

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions
Variant 2 (Higher Difficulty)

Time: 50 minutes | Total marks: 30

Topic focus: Thermal physics, conduction, energy transfer, efficiency, phase change, and experimental evaluation.

Data and constants

- Area of solar collector panel, $A = 4.0\,\mathrm{m}^2$
- Average solar radiation intensity, $I = 780\,\mathrm{Wm}^{-2}$
- Collector optical-thermal efficiency at start of test, $\eta_0 = 0.68$
- Water mass in storage tank, $m_w = 220\,\mathrm{kg}$
- Initial tank temperature, $T_i = 16.0\,°\mathrm{C}$
- Ambient temperature, $T_{air} = 12.0\,°\mathrm{C}$
- Tank surface area, $A_{tank} = 2.8\,\mathrm{m}^2$
- Insulation thickness, $d = 4.0\,\mathrm{cm}$
- Insulation thermal conductivity, $k = 0.030\,\mathrm{Wm}^{-1}\mathrm{K}^{-1}$
- Specific heat capacity of water, $c_w = 4200\,\mathrm{Jkg}^{-1}\mathrm{K}^{-1}$
- Specific latent heat of fusion of ice, $L_f = 3.34 \times 10^5\,\mathrm{Jkg}^{-1}$
- Specific latent heat of vaporisation of water, $L_v = 2.26 \times 10^6\,\mathrm{Jkg}^{-1}$

Additional model assumptions for Variant 2

- Test duration: $t = 6.0\,\mathrm{h}$.
- Due to increasing collector temperature, efficiency decreases linearly from 0.68 to 0.56 over the 6.0 h interval.
- Mean water temperature during loss calculations may be taken as $55.0\,°\mathrm{C}$, unless otherwise stated.

Question 7 (Variant 2)

(a) Calculate the incident solar power on the panel and the total incident solar energy over 6.0 h. [3]

(b) Determine the useful thermal energy gained by the water by using the *mean efficiency* over the interval. [3]

(c) Ignoring all thermal losses from the tank, calculate the predicted final water temperature. [3]

(d) Using conduction only through the insulation,

(i) calculate the rate of heat loss at mean water temperature 55.0 °C;

(ii) calculate the total energy lost by conduction in 6.0 h.

[4]

(e) Calculate a more realistic final tank temperature including the conduction loss from (d). [3]

(f) A 1.5 kg sample of water from the tank at temperature found in (e) is mixed with 0.65 kg of ice at 0 °C in an insulated container.

Determine whether all the ice melts. If all ice melts, calculate the final equilibrium temperature of the mixture. [5]

(g) Open-ended conceptual question

The heating system is to be redesigned for a colder high-altitude site where midday irradiance fluctuates rapidly due to intermittent cloud cover.

Propose and justify **three** physics-based modifications that would significantly improve **daily delivered hot-water energy** while keeping the same collector area.

For each modification, your answer should:

- identify the dominant physical process affected,
- predict the direction of change in performance,
- include one possible trade-off or practical limitation.

[6]

(h) Open-ended evaluation question

Design an experiment to determine how the effective heat-loss coefficient of the tank depends on water temperature.

Your plan should include:

- measured variables and apparatus,
- method to control key variables,

- data treatment leading to a graph and extraction of a coefficient,
- two significant uncertainty sources and how to reduce them.

[3]

Sample Solution and Mark Scheme (CIE P4 style) – Variant 2 only

Total: 30 marks

(a) [3]

Incident power:

$$P_{in} = IA = (780)(4.0) = 3120\,\mathrm{W}$$

Time:

$$t = 6.0 \times 3600 = 21\,600\,\mathrm{s}$$

Incident energy:

$$E_{in} = P_{in}t = (3120)(21600) = 6.7392 \times 10^7\,\mathrm{J}$$

Mark scheme: $M1$ use $P = IA$; $A1$ $3120\,\mathrm{W}$; $A1$ correct energy $6.74 \times 10^7\,\mathrm{J}$ (allow 3 s.f.).

(b) [3]

Mean efficiency:

$$\bar{\eta} = \frac{0.68 + 0.56}{2} = 0.62$$

Useful gain:

$$Q_{gain} = \bar{\eta}E_{in} = 0.62(6.7392 \times 10^7) = 4.1783 \times 10^7\,\mathrm{J}$$

Mark scheme: $B1$ mean efficiency; $M1$ multiply by incident energy; $A1$ $4.18 \times 10^7\,\mathrm{J}$.

(c) [3]

Ignoring losses:

$$Q_{gain} = m_w c_w (T_f - T_i)$$

$$T_f = 16.0 + \frac{4.1783 \times 10^7}{(220)(4200)} = 16.0 + 45.2 = 61.2\,^\circ\mathrm{C}$$

Mark scheme: $M1$ correct thermal equation; $M1$ substitution; $A1$ $61.2\,^\circ\mathrm{C}$.

(d) [4]

(i) Temperature difference:

$$\Delta T = 55.0 - 12.0 = 43.0\,\mathrm{K}$$

Conduction loss rate:

$$P_{loss} = \frac{kA_{tank}\Delta T}{d} = \frac{(0.030)(2.8)(43.0)}{0.040} = 90.3\,\mathrm{W}$$

(ii) Energy loss in 6.0 h:

$$Q_{loss} = P_{loss}t = (90.3)(21600) = 1.95 \times 10^6 \text{ J}$$

Mark scheme: $B1$ $\Delta T = 43$ K; $M1$ correct conduction formula/use of $d = 0.040$ m; $A1$ 90.3 W; $A1$ 1.95×10^6 J.

(e) [3]

Net gain:

$$Q_{net} = Q_{gain} - Q_{loss} = 4.1783 \times 10^7 - 1.95 \times 10^6 = 3.98 \times 10^7 \text{ J}$$

$$T_f = 16.0 + \frac{Q_{net}}{m_w c_w} = 16.0 + \frac{3.98 \times 10^7}{(220)(4200)} = 59.1\,^\circ\text{C}$$

Mark scheme: $M1$ net energy subtraction; $M1$ use $Q = mc\Delta T$; $A1$ 59.1 °C (accept 59–60 °C).

(f) [5]

Energy available from warm water cooling to 0 °C:

$$Q_w = (1.5)(4200)(59.1) = 3.72 \times 10^5 \text{ J}$$

Energy to melt all ice:

$$Q_m = (0.65)(3.34 \times 10^5) = 2.17 \times 10^5 \text{ J}$$

Since $Q_w > Q_m$, all ice melts.

Remaining energy after melting:

$$Q_{rem} = 3.72 \times 10^5 - 2.17 \times 10^5 = 1.55 \times 10^5 \text{ J}$$

Total liquid mass after melting: $m_{tot} = 1.5 + 0.65 = 2.15$ kg.

$$T_{eq} = \frac{Q_{rem}}{m_{tot} c_w} = \frac{1.55 \times 10^5}{(2.15)(4200)} = 17.2\,^\circ\text{C}$$

Mark scheme: $M1$ hot-water energy expression; $A1$ sensible value; $M1$ compare with latent requirement; $A1$ conclusion "all ice melts"; $A1$ final temperature 17.2 °C (allow 16–18 °C).

(g) Open-ended conceptual [6]

Indicative sample response (any equivalent physics arguments acceptable):

- **Selective absorber coating (low emissivity, high absorptivity).** Increases absorption of short-wave solar radiation while reducing long-wave thermal emission from the collector. Net gain increases, especially at high plate temperature. Trade-off: higher material cost and possible coating degradation.

- **Improved tank insulation and thermal-bridge reduction.** Increase insulation thickness and eliminate metal conduction paths at supports/pipe joints to reduce conductive and mixed losses. This improves overnight energy retention. Trade-off: larger tank volume and possible maintenance access issues.
- **Variable-speed pumped circulation with differential temperature control.** Pump runs only when collector temperature exceeds tank temperature by a threshold, limiting reverse heat flow during cloud events. Raises daily useful efficiency under fluctuating irradiance. Trade-off: control complexity and electrical parasitic load.

Mark scheme: award up to 2 marks per valid modification: 1 mark for correct physics mechanism + 1 mark for justified performance/trade-off discussion, max 6.

(h) Open-ended evaluation [3]

Indicative sample response:

- Heat water to a chosen initial temperature, isolate tank from solar input, record $T(t)$ and ambient temperature with calibrated sensors and data logger.
- Repeat for several starting temperatures while keeping airflow, exposed area and insulation state constant.
- Use $P_{loss} = mc\left|\frac{dT}{dt}\right|$ and plot P_{loss} against $(T - T_{air})$; gradient gives effective conductance (or convert to coefficient per area). Main uncertainties: sensor calibration/lag and uncontrolled convection; reduce via sensor calibration, shielding from drafts, and repeated runs.

Mark scheme: $B1$ suitable measurable method; $B1$ correct analysis/graph to extract coefficient; $B1$ valid uncertainty discussion with mitigation.